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iSense

DATA SHEET

PRODUCT INFO

PROCESSOR
MediaTek MT7621, 512MB DDR, 32MB/64MB Flash.

CELLULAR
Single / Dual SIM with Quad Band CAT4/CAT6/CAT12 GSM/GPRS/GPS/2G/3G/4G/5G [M.2 Based]

Wi-Fi
Dual-band Wi-Fi (IEEE802.1 b/g/n/ac, 2.4GHz/5GHz)

INTERFACE
1 WAN & 4 LAN 10 /100 / 1000 Mbps, 1 RS232 port [DB-9], 2 X Isolated RS485 (Half Duplex) port [3 pin terminal]

POWER
20 to 60 Volts DC (48V nominal) [Modular]

OPERATING SYSTEM
Linux

Hardware Specification

CPU	MediaTek MT7621
RAM	512MB
FLASH	32 / 64 MB Flash
Ethernet	4x 10/100/1000 (Configurable WAN/LAN)
USB	1 x USB 3.0 Host
Serial Interface	
• Digital Inputs	12V / 24V with isolation, 12 Digital Inputs (External/self powered)
• Digital outputs	4 Potential Free Outputs (SPDT Relay)
• Temperature Sensor	2 External Temperature Sensor Probes
• Analog Input (Voltage)	2 with Isolation (Range: -60 to 60V)
• Analog Input (Current)	2 with isolation (Range: 4-20mA)
• RS485	4 X Isolated RS485 (Half Duplex) port [Modular]
• RS232	1 RS232 port
SIM	2x Nano SIM (1x Active & 1x Passive)
Wi-Fi	802.11 b/g/n/ac
RTC	Onboard RTC with battery backup
Reset	Pinhole Reset Button
Antenna	Cellular: 2x SMA Connector, Wi-Fi: 4x SMA connector
Status LED	On-board LED for status indication(Power, Cellular ACT, Wi-Fi, Programmable LED etc)
Power Connector	2-pin Screw Terminal connector
Input Voltage Range	9-30 Vdc
Enclosure	TBD Sheet Metal with Powder coating
Mounting	Rack Mount (45 mm Depth)
Design	Fanless
Dimensions	290 x 38 x 124 mm (W x H x D)
Weight	1856 gm
Other	Grounding Screw

Operational & Environmental

Storage Temperature	-40°C to 85°C
Operating Temperature	-10°C to 70°C
Humidity	90% non-condensing

SOFTWARE SPECIFICATIONS

Cellular

Cellular Module	4G (LTE) - Cat 4 up to 150 Mbps, 3G - up to 42 Mbps, 2G - up to 236.8 Kbps (Optional 5G)
Status	Single strength (RSSI), SINR, RSRP, RSRQ
Band Management	Auto/User configurable
Operator Lock	Automatic/Manual
APN	Configurable, Auto APN
Modem configuration mode	Single SIM, Dual SIM
SIM switching	Signal strength, ICMP echo, Data limit based
PDP Type	IPv4, IPv6, IPv4v6
SIM Authentication	PAP,CHAP, No Authentication
SMS	Send SMS, Read SMS, SMS Commands

Firewall & Networking

Routing	Static routing, Advanced Static Routing, Dynamic Routing
Network Protocols	TCP, UDP, IPv4, IPv6, ICMP, NTP, HTTP, HTTPS, SCP, PPP, SSH, DHCP
Connection Monitoring	Ping Reboot, Scheduled Periodic Reboot
Firewall features	Port forward, traffic rules, SNAT, Zone forwarding, Parental Control, Masquerading, MSS Clamping, Loopback IP, Loopback Rules.
Attack Prevention	SYN Flood Protection, Drop Invalid Packets
DHCP	Static and Dynamic
Multi-WAN Configuration	Configurable Failover, load balancing between Cellular WAN, Wi-Fi WAN and Wired.
VLAN	Tagged VLAN, Untagged VLAN
VPN	Open VPN, IPsec, GRE, L2TP, PPTP, Wireguard, Zero Tier, IPIP Tunnel
Open VPN	TUN/TAP with TLS Certificates
IPsec	IKEv1, IKEv2, Aggressive/Main Mode
IPsec Encryption	3DES, AES128, AES192, AES256
IPsec Authentication	SHA1, MD5, SHA256, SHA384, SHA512
IPsec DH Group	MODP768, MODP1024, MODP1536, MODP2048, MODP3072, MODP4096
PPTP/L2TP	Server & Client with/without MPPE Encryption
Monitoring Application	Modem Monitoring with ICMP Echo, Router Monitoring with ICMP Echo
Time Synchronization	NTP Client, RTC, Manual
Diagnostic	Ping, Traceroute, Modem Debug Info, Iperf, Download Logs
Other Features	MAC binding, Connection Diagnostics, URL Filtering, Router Health Monitoring, Package Manager

Remote Management

NMS	Remote Monitoring & Management Platform for SILBO Networking products
NMS VPN	Remote Access of SILBO and non-SILBO Networking Devices
TR069	TR069 Client for CPE Management
SMS Commands	Reboot Command, Uptime Command

Gateway

Gateway Protocols	Modbus TCP Slave, Modbus TCP Master
Serial Mode	Transparent (Over IP), Modbus Gateway
Data sender	SNMP, HTTP, MQTT
Internal Storage	4GB eMMC(optional)

Operating System

Operating System	Open-WRT based Linux OS
Supported Languages	Shell scripting, C, C++
Development tools	SDK package with built environment provided (Optional)

Standard Packaging

iSense Cellular Gateway with Sheet Metal Housing, 4x Wi-Fi Rubber Duck Antennas, 2x Cellular Rubber Duck Antennas, 0.5-meter Power Cord, 01-meter Ethernet cable, Serial interface connectors.

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